

INSTITUTE OF ACCOUNTANCY ARUSHA



INDIVIDUAL ASSIGNMENT

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PROGRAM: BACHELOR IN COMPUTER SCIENCE

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Introduction

Cybersecurity is a very important aspect of modern digital life. With the rapid growth of internet usage, individuals and organizations are increasingly exposed to cyber threats. Cyberattacks are malicious attempts to access, damage, or disrupt systems and data. Understanding common types of cyberattacks and how to prevent them is critical in maintaining security.

• Phishing Attacks

Phishing is a cyberattack where attackers impersonate legitimate organizations to trick users into revealing sensitive information such as passwords or bank details. It is one of the most common cyberattacks. Examples of Phishing Attacks are fake bank emails requesting account verification, messages claiming that a user has won a prize, fake login pages that steal usernames and passwords.

Prevention:

- Avoid clicking suspicious links or attachments
- Verify email sources before responding
- Use spam filters as well as email security software
- Educate users on identifying Phishing Attacks -enable multifactor authentication.

• Malware Attacks

These are malicious software designed to damage or gain unauthorized access to computer systems. Malware includes viruses, worms, and trojans that harm systems or steal data.

Types of malwares

‡ Virus

A virus attaches itself to files and spreads when the file is opened.

‡ Worm

A worm spreads automatically across networks without user interaction. ‡

Spyware

It secretly monitors user activities and steals information.

‡ Trojan horse

It disguises itself as legitimate software but contains harmful code.

Prevention:

- Install antivirus software
- Avoid unsafe downloads of software from untrusted websites
- Update systems regularly
- Use firewalls so as to block unauthorized access

• **Ransomware attacks**

Ransomware locks user data and demands payment to restore access. attackers usually demands payment in cryptocurrency.

Effects of Ransomware attacks

- Loss of access to important files
- Financial losses
- Business interruption
- Data leakage **Prevention:**
- Backup important data regularly
- Avoid suspicious attachments
- Use updated security tool

- Use endpoint protection software
- Train employees about ransomware threats

- **DoS and DDoS Attacks**

Here attackers flood the system or network with excessive traffic making it unavailable to users.

Effects

- Website downtime
- Slow network performance
- Service interruption

Prevention:

- Use firewalls and intrusion detection system(IDS)
- Monitor network traffic continuously
- Use DDoS protection services
- Increase network bandwidth and server capacity

- **Man-in-the-Middle Attacks**

It occurs when Attacker intercept communication between two parties. Examples intercepting data on public wifi networks, stealing login credentials during online transactions.

Prevention:

- Use HTTPS

- Avoid the use of public Wi-Fi
- Enable secure communication protocols.
- Use VPN

- **SQL Injection**

Occurs when Attackers insert malicious SQL code into website forms or applications to access or manipulate databases through input fields.

Effects

- Theft of sensitive data
- Database corruption
- Unauthorized access

Prevention:

- Validate user input properly
- Use secure coding practices
- Use parameterized queries -Update database systems regularly
- Limit database access ermissions.

- **Password Attacks**

Occurs when Attackers try to gain access to accounts by guessing or stealing passwords. Common passwords attacks include brute force attack, dictionary attack and credential stuffing.

Prevention:

- Use strong and unique passwords
- Enable multi-Factor Authentication (MFA)
- Limit login attempts -Avoid passwords sharing.
- Use passwords managers.

Generally, cyberattacks continue to increase as technology advances which cause serious risks to individuals and organizations. These cyber-attacks lead to financial losses, data breaches, and disruption of services. However, many of these cyber attacks can be prevented through proper cyber security practices including user education, strong passwords, regular updates, antivirus protection, and secure network usage.

Organizations as well as individuals must remain vigilant and proactive in implementing security measures to protect the systems and information from cyber threats.

Cyber security is not only the responsibility of IT professionals but also every computer user.

References

Vectra AI & SentinelOne Reports on IoT Security (2025-2026)

IoT Analytics – Global Connected Devices Report

Asimily – Major IoT Cybersecurity Breaches

Various industry articles and academic sources on IoT risks

Word Count: Approximately 2,450 words

This report is now longer and more presentable with a diagram and a clear risk table. You can copy it into Microsoft Word, adjust the formatting (1.5 line spacing, 12pt font), and add the diagram as a proper drawn figure if you want.